Travel Mode Prediction

Using 2017 NHTS Data & Machine Learning Classification

DSCI 510 · Spring 2026 · University of Southern California

Ming Xia

Introduction

Objective

Predict individual trip transportation mode (TRPTRANS) using household, person, and trip-level features from NHTS 2017.

Dataset

2017 National Household Travel Survey — 923,572 trip records merged with household & person data.

Approach

Feature engineering → One-hot encoding → Train & compare Logistic Regression, Decision Tree, and XGBoost classifiers.

20

Transportation Modes

31

Features Selected

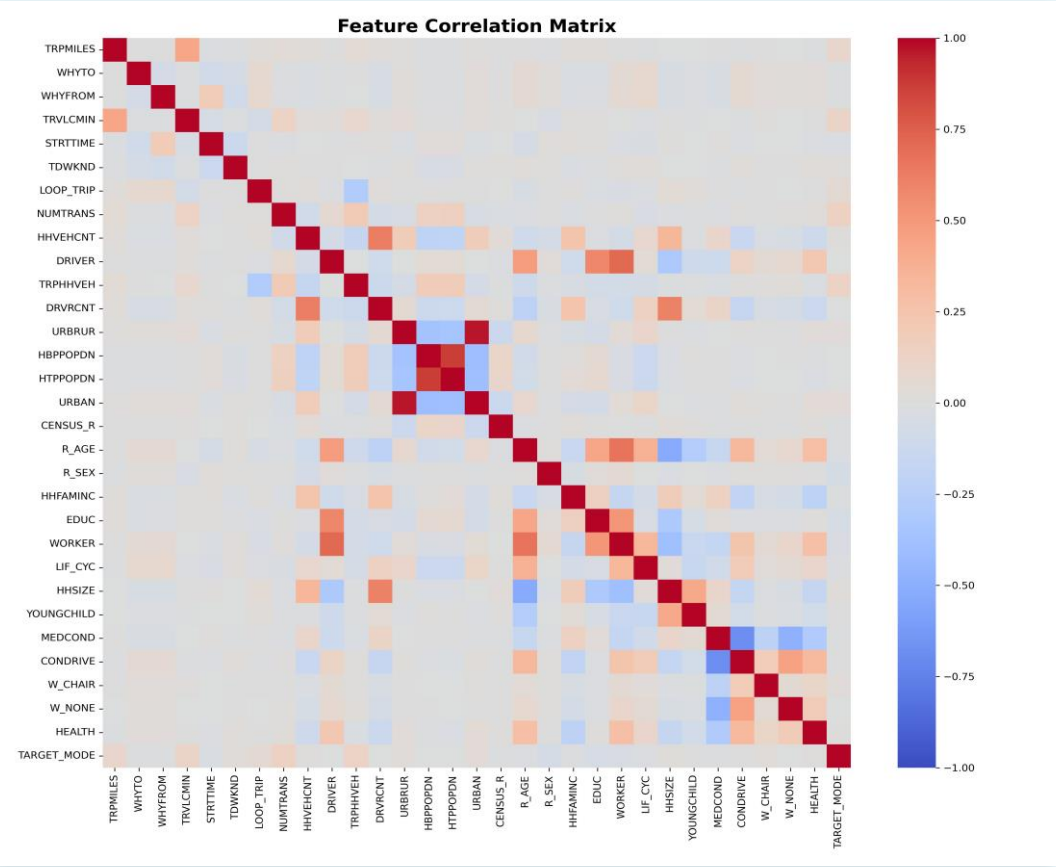
3

ML Models Compared

Data Sources

Source	Description	Records	Key Variables
NHTS 2017 trippub.csv	Trip-level travel survey data	923,572 trips	TRPTRANS, TRPMILES, TRVLCMIN, WHYTO
NHTS 2017 perpub.csv	Person-level demographics	264,234 persons	R_AGE, R_SEX, EDUC, MEDCOND
NHTS 2017 hhpub.csv	Household-level attributes	129,696 households	HHVEHCNT, HHFAMINC, URBRUR
US Census ACS 2024	Income & socio-economic data (API)	National level	S1901 income table

Results — Feature Correlation Analysis



Strong multicollinearity

URBRUR / HBPPOPDN / HTPPOPDN / URBAN represent the same urbanization signal.

Vehicle features cluster

HHVEHCNT, DRIVER, DRVRCNT and TRPHHVEH form a tight group; all strong predictors.

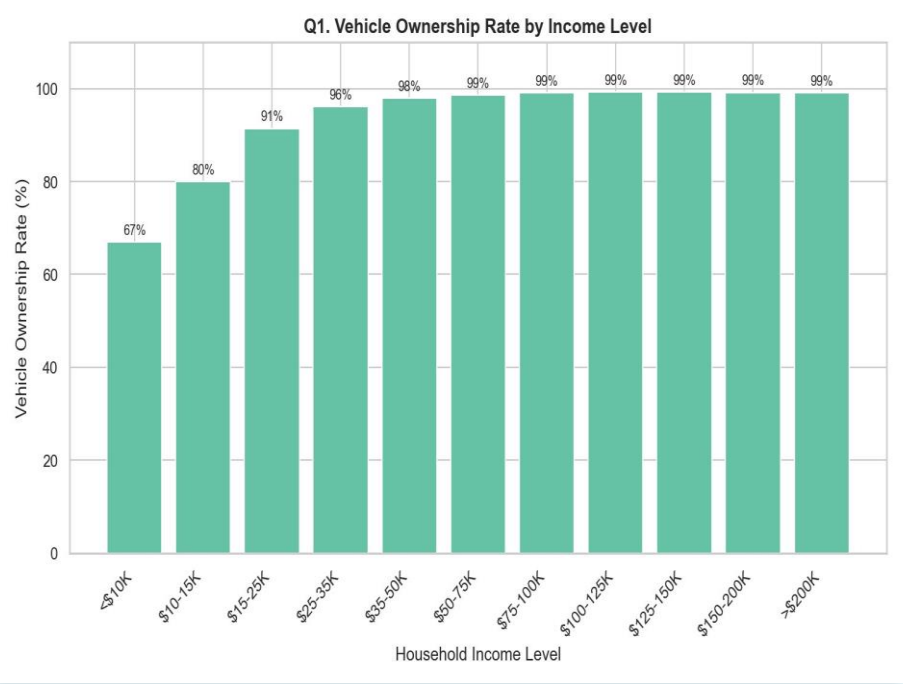
Health variables overlap

MEDCOND, CONDRIVE, W_CHAIR and W_NONE are inter-correlated — candidates for reduction.

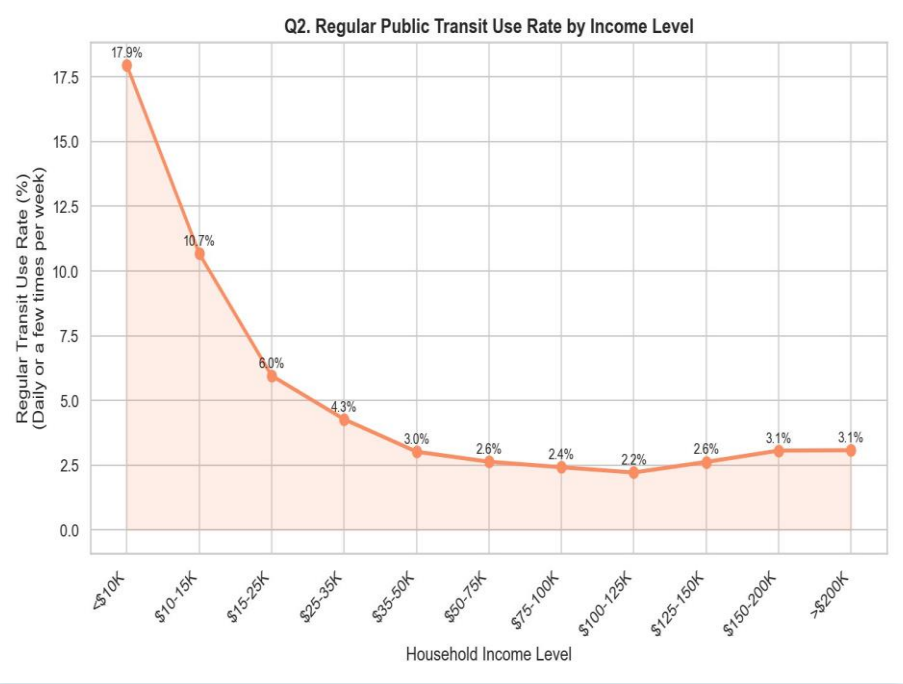
Weak linear signal to target

Near-zero correlation with TARGET_MODE confirms non-linear models are necessary.

Exploratory Analysis — Income & Vehicle / Transit Use



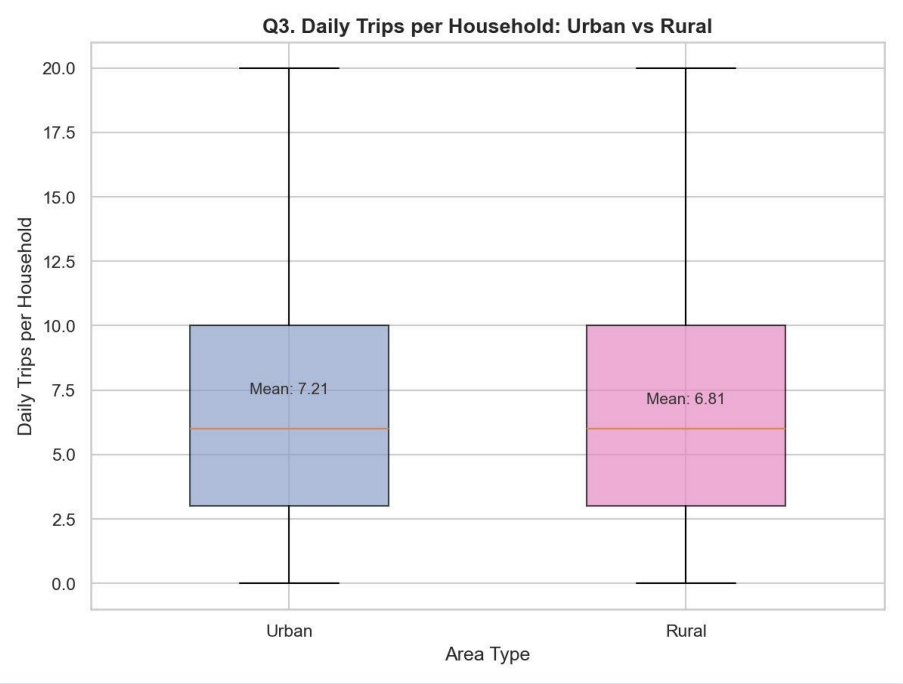
Q1. Vehicle Ownership Rate by Income Level



Q2. Regular Public Transit Use Rate by Income Level

Finding: Vehicle ownership rises sharply with income (67% → 99%). Conversely, low-income households rely on public transit far more (17.9% vs ~2.5%). Income is a key socioeconomic driver of travel mode.

Exploratory Analysis — Urban/Rural & Walking by Age

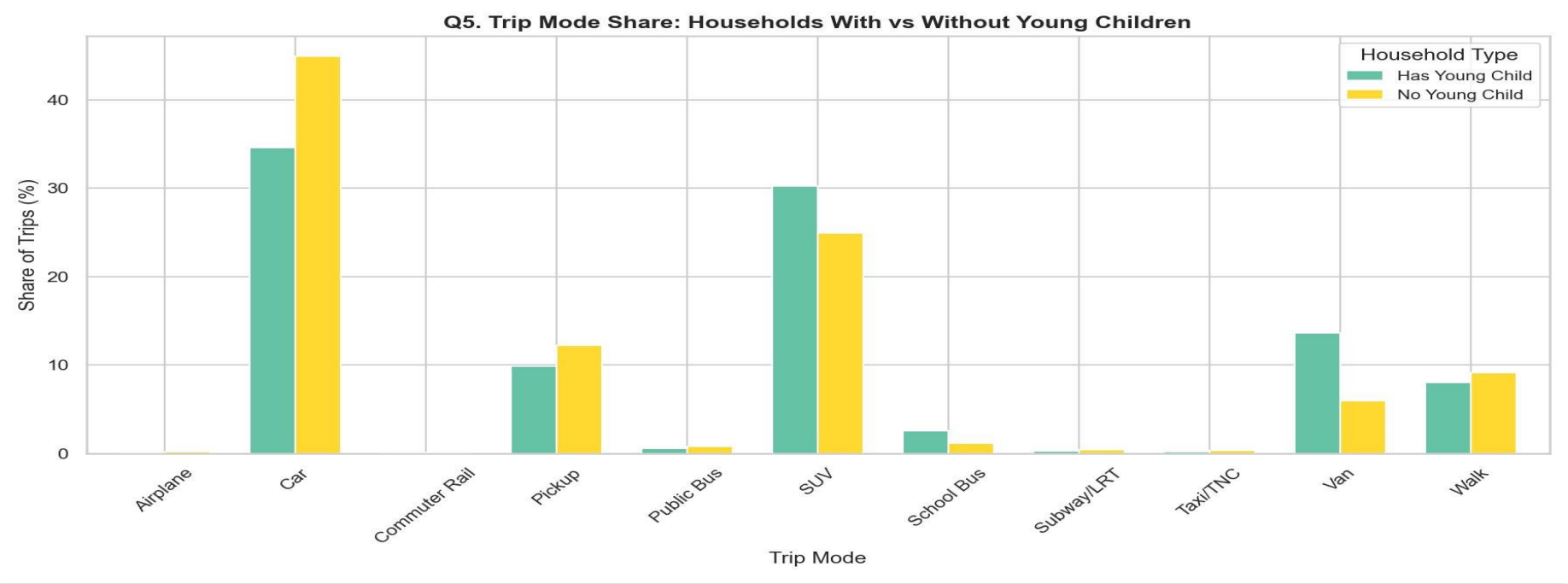


Q3. Daily Trips: Urban vs Rural

Q4. Regular Walking Rate by Age Group

Finding: Urban households make slightly more daily trips (7.21 vs 6.81). Older adults (60+) walk most regularly (38%+), while minors walk least (31.3%) — suggesting retirement frees up time for active travel.

Exploratory Analysis — Children & Urbanization

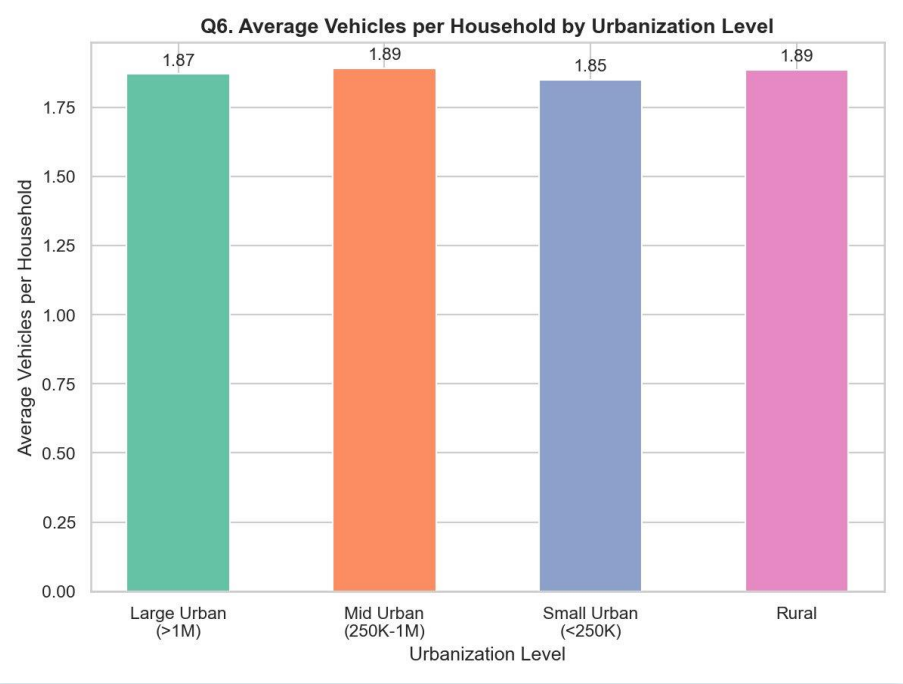


Q5. Trip Mode Share: Households With vs Without Young Children

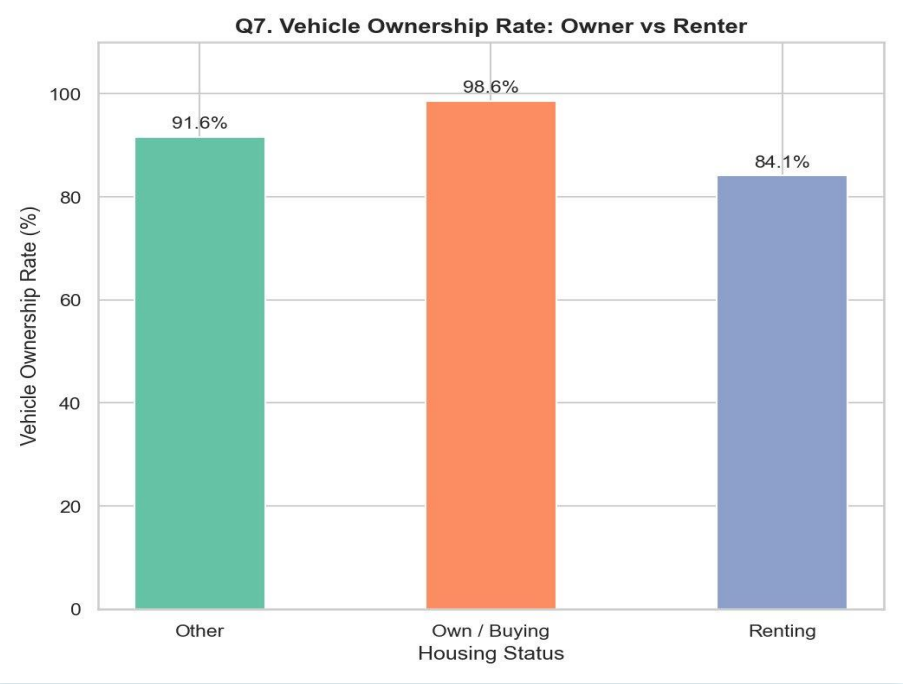
Q5: Families with young children use SUVs (30%) and Vans (14%) more; school bus only appears in child households.

Q6: Average vehicles per household is nearly identical across all urbanization levels (1.85–1.89) — cars are a universal necessity in the US.

Exploratory Analysis — Urbanization & Housing Status



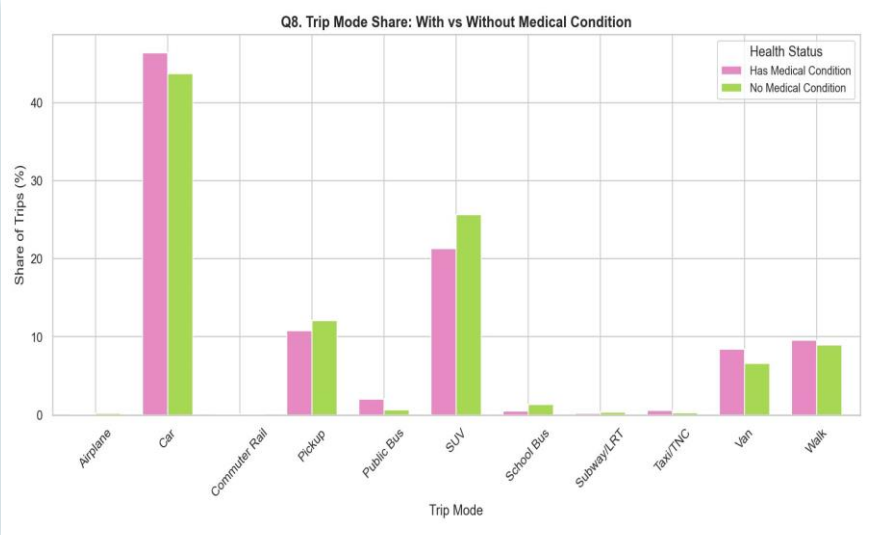
Q6. Avg Vehicles per Household by Urbanization Level



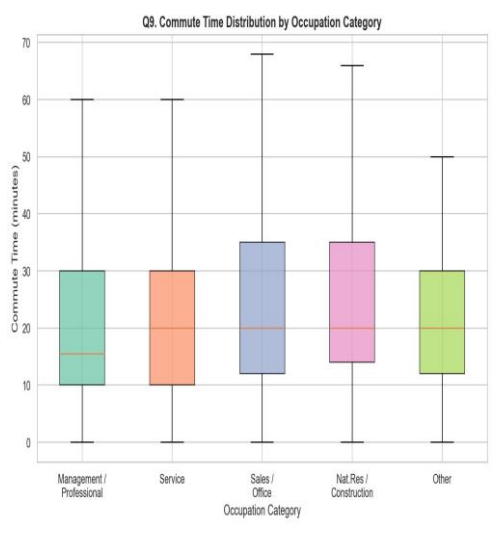
Q7. Vehicle Ownership Rate: Owner vs Renter

Finding: Vehicle count is uniform across urban/rural areas (~1.87). However, homeowners have a significantly higher ownership rate (98.6%) vs renters (84.1%), reflecting a 14.5-point wealth gap.

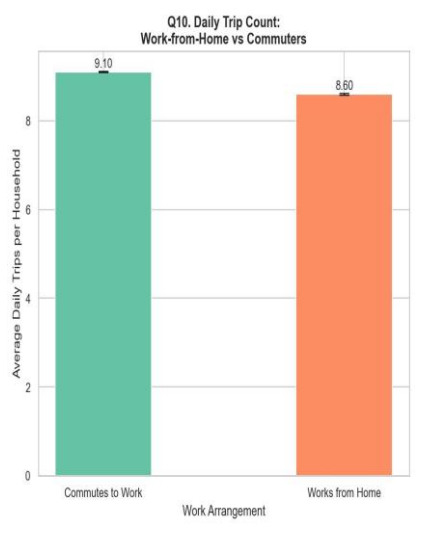
Exploratory Analysis — Health, Occupation & Work Arrangement



Q8. Mode Share by Medical Condition



Q9. Commute Time by Occupation



Q10. WFH vs Commuters

Q8 — Medical Condition

People with medical conditions rely more on cars (46%) and vans; higher public bus use reflects accessibility needs.

Q9 — Occupation

Median commute ~20 min across all jobs. Sales/Office and Construction show wider spread — less predictable locations.

Q10 — Work Arrangement

Commuters average 9.10 daily trips vs 8.60 for WFH workers. Removing commute only reduces total by ~0.5 trips.

Results — Model Performance Comparison

54.22%

Logistic Regression

- Baseline linear model
- Lowest accuracy
- Struggles with class imbalance & non-linearity

57.94%

XGBoost

- Ensemble gradient boosting
- Middle accuracy
- Expected best — may need further tuning

61.32%

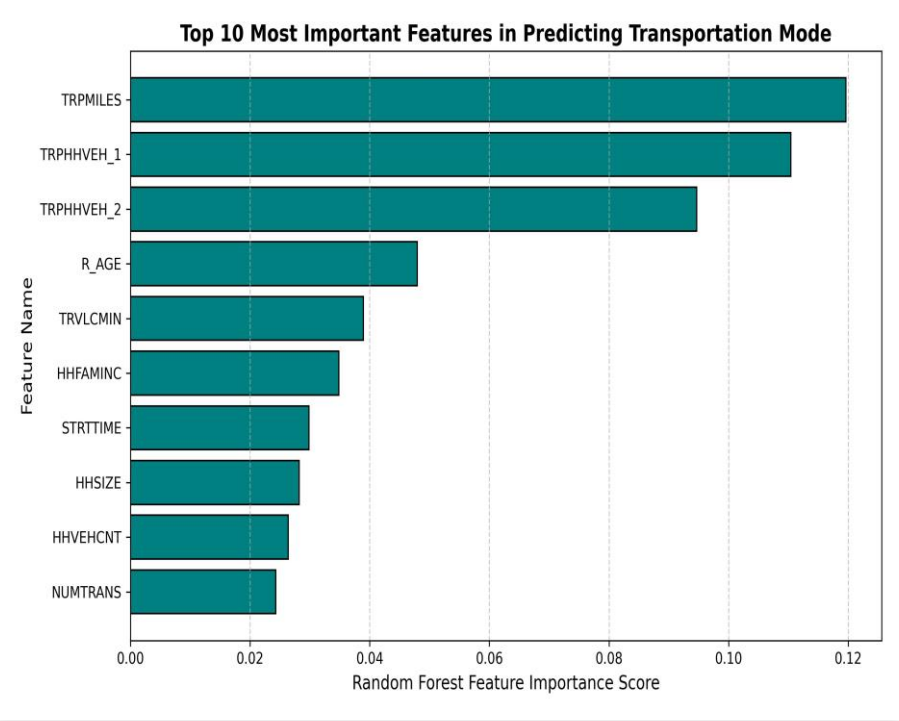
Decision Tree

- Max depth = 20
- Highest accuracy in this run
- Likely capturing majority class well

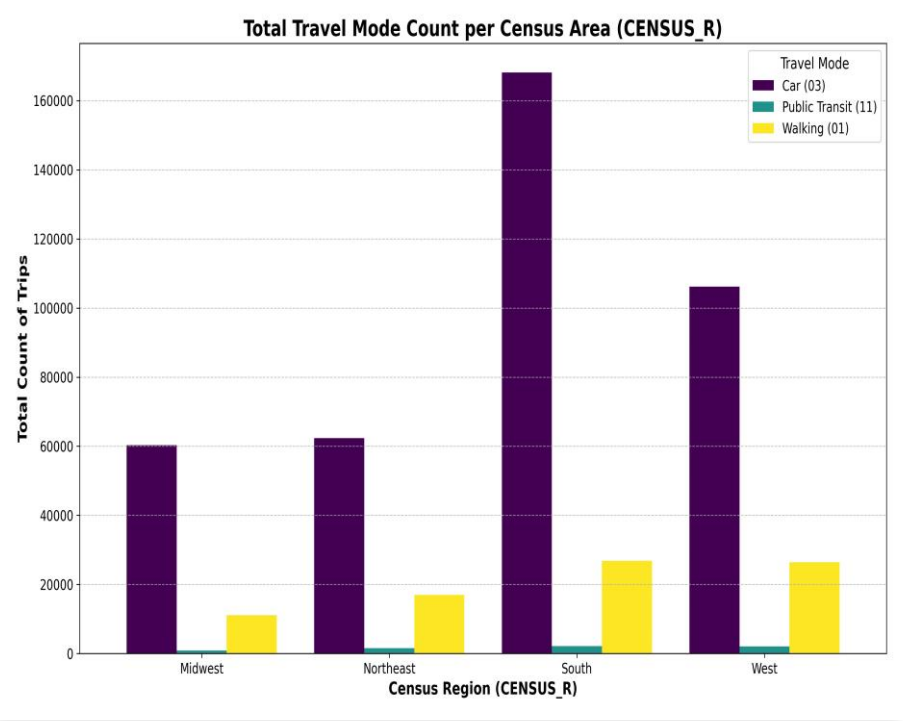
★ Highest Accuracy

⚠ Note: Overall accuracy (54–61%) is moderate due to class imbalance — personal vehicle trips dominate at 80%+. F1-score per class would better reflect true model performance.

Results — Feature Importance & Regional Travel Patterns



Top 10 Most Important Features (Decision Tree)



Travel Mode Count by Census Region

Key finding: TRPMILES (trip distance) is the single most important predictor. Car dominates all regions; the South has the highest trip volume.

Challenges

01 Class Imbalance

Personal vehicle trips dominate at 80%+, which likely explains the moderate accuracy (54–61%). Stratified splitting was applied; further work with SMOTE or class weights may improve minority-class prediction.

02 Memory & Scale

Merging 923K trip records with household and person data creates a very large one-to-many join. Required column filtering and chunked processing to stay within memory limits.

03 Feature Selection

NHTS codebook contains 200+ variables. Identifying the top 31 features required reading the codebook, correlation analysis, and Random Forest importance scoring.

04 Negative Value Encoding

NHTS uses codes −1, −7, −8, −9 for missing/refused values — not true negatives. Required careful filtering before any numeric imputation or model training.